MILITARY SPECIFICATION

CLOTH, SERGE: POLYESTER/WOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of cloth, serge, polyester/wool.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMA-1), Headquarters U. S. Marine Corps, Washington, D. C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-655 - Provision for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

LAWS AND REGULATIONS

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

U. S. Department of Agriculture

Methods of Test for Grades of Wool Top

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Marketing Service, Washington, D.C. 20402.)

Technical Manual of the American Association of Textile Chemists and Colorists

Method No. 128-1969 - Wrinkle Recovery of Fabrics Appearance Method

(Application for copies should be addressed to the AATCC, P. O. Box 12215, Research Triangle Park, N. C. 27709.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample cloth for first article inspection and approval. The first article shall consist of 5 yards, full width. The approval of the first article authorizes the commencement of production, but does not relieve the contractor of the responsibility for compliance with

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all provisions of the specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production cloth.

3.3 Material.

3.3.1 Fibers.

3.3.1.1 Wool. The stock shall be fleece or pulled wool or both, not lower in grade than 62's U. S. Standard, when tested as specified in 4.5.1.2. The use of noils, laps, or other wool manufacturing by-products is prohibited.

3.3.1.2 Polyester. The stock shall be semidull polyester fiber made from polyethylene glycol terephthalate, either homopolymer or modified polymer as appropriate (see 4.5.1 and 6.5). The minimum average spun fiber length shall be 3-1/4 inches when tested as specified in 4.5.1.1. The continuous filament textured yarn shall be 2/150/34 denier. The staple spun polyester shall be 3.0 denier. The use of polyester fiber waste is prohibited.

3.3.2 Yarns. The spun polyester and wool fiber shall be blended into a single spun yarn and woven with a 100 percent continuous filament textured polyester yarn into an end and end, pick and pick arrangement in both warp and filling direction. The finished cloth shall consist of the following fiber percentages based on the oven-dry weight when tested as specified in 4.7.

77 percent (min), 83 percent (max) polyester fiber,

17 percent (min), wool fiber, 62's U. S. Standard (or finer)

NOTE: Selvage yarn shall be the same as the 100 percent continuous filament textured polyester yarn used in the body of the cloth and will be 2/150/34 denier (see 3.11).

3.4 Design and construction.

3.4.1 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.7.

Table I. Physical requirements

Weight per Linear yd. 56 inches		Yarns per inch Minimum		Breaking strength Minimum		Tearing strength Minimum	
Min.	Max.	Warp	Filling	Warp	Filling	Warp	Filling
Ounces				Pounds		Pounds	
10.5	11.0	65	55	160	140	6.0	6.0

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3.4.2 Weave. The weave shall be a two up and two down right hand twill.

3.5 Performance.

3.5.1 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.5 percent in the direction of the warp or filling when tested as specified in 4.7.

3.5.2 pH. The pH value of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.7.

3.6 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 50 yards. Each length shall be put-up on full width rolls as specified in PPP-P-1132.

3.7 Width. Unless otherwise specified (see 6.2), the width shall be no less than 60 inches exclusive of selvages or exclusive of tuck in when woven on sulzer-type looms.

3.8 Color. The color shall be as specified (see 6.2 and 6.4) and shall match the standard sample. The color may be obtained by piece dyeing. Monotone shades regardless of blend composition shall present a solid appearance with no more heatheriness than is displayed by the standard sample.

3.8.1 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamp light at 2800° Kelvin.

3.8.2 Colorfastness. The finished cloth shall show fastness to wet (dry) cleaning, crocking, perspiration and light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to wet (dry) cleaning, perspiration, and light, and shall show a Munsell Value for crocking not less than 3.0. The fastness to light exposure time for good fastness shall be 40 standard fading hours. Test shall be as specified in 4.7.

3.9 Mothproofing. Mothproofing of the wool material in accordance with MIL-C-43665 shall be accomplished during dyeing as applicable (see 4.7).

3.10 Finish. The cloth shall be scoured, brushed, sheared, pressed and decated so as to provide a finish equal to that of the standard sample. The finished cloth shall show no more creping after shrinkage

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test than exhibited by the standard sample. The finished cloth shall not show wrinkling to a degree greater than as shown by the standard sample after one dry cleaning. Testing shall be as specified in 4.7.

3.10.1 Pilling. The cloth shall be singed and finished to show no more pilling than the standard sample when tested as specified in 4.7 (see 6.5.4)

3.11 Selvages. The selvages shall be so constructed that in the finished cloth they shall lie flat when the material is laid out for cutting.

3.12 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than 1/2 inch in height (see 4.5.2.3). The marking shall be clearly legible in any indelible marking medium used commercially.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1-1/2 inch long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string represent a one or two point defect (see 4.5.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. When first article is required, it shall be tested as specified in 4.7.

4.5 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-655, except where otherwise indicated hereinafter.

4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.5.1.1 Polyester. The polyester fiber shall be tested and identified as specified in Table II.

Table II. Component test method

Characteristic	Requirement paragraph	Test method
Polyester identification	3.3.1.2	4.7.1 1/
Length of fiber	3.3.1.2	Visual 1/
Denier	3.3.2	4021 1/

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.5.1.2 Wool. The wool shall be examined for grade (see 3.3.1.1) in undyed top form by comparison with the applicable U. S. Standard (see 2.1). In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be 1 yard of undyed top. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 pound. The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade.

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.5.1.3 In-process inspection. Inspection shall be made at any point, or during any phase, of the manufacturing process to assure that only approved components are used and that the dyeing requirements have been adhered to (see 3.8 and 3.10.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

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4.5.2 Examination of the end item. Examination of the end item shall be in accordance with 4.5.2.1 through 4.5.2.5.

4.5.2.1 Yard-by-yard examination. The yard-by-yard examination shall be as specified in MIL-STD-655. The fabric quality for all shades shall be grade 15.

4.5.2.1.1 Limits of slubs and knots. Only slubs and knots which exceed "D" for slubs and "C" for knots on the Sears Fabric Defect Scale (see 6.6) shall be scored.

4.5.2.2 Examination for length. The examination for length shall be as specified in MIL-STD-655.

4.5.2.3 Examination for back marking and compliance with the Wool Products Labeling Act. The examination for back marking and compliance with the Wool Products Labeling Act (see 2.1) shall be as specified in MIL-STD-655 .

4.5.2.4 Examination for shade match and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.8) and finish (see 3.10.) A roll shall be unacceptable if it fails to meet the requirements for shade match or finish. The sample unit shall be a 4 inch by 20 inch swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.5.2.5 Examination for defect marking (stringing of defects). The examination for defect marking shall be specified in MIL-STD-655.

4.6 Examination of preparation for delivery requirements. Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the Section 5 requirements.

4.7 Tests. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table III shall be followed. The physical and chemical values specified in Section 3 apply to the result of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

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Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table III. Test methods

Characteristic	Requirement paragraph	Test method
Yarn single or ply	3.3.2	Visual
Fiber content:		
Polyester (percent)	3.3.2	2102
Wool (percent)	3.3.2	2102
Mothproofing content	3.9	1/
Colorfastness:		
Wet (dry) cleaning	3.8.2	5622
Crocking	3.8.2	5651
Light	3.8.2	5660
Perspiration	3.8.2	5680
Weight	3.4.1	5041
Yarns per inch:		
Warp	3.4.1	5050
Filling	3.4.1	5050
Breaking strength:		
Warp	3.4.1	5100
Filling	3.4.1	5100
Tearing strength:		
Warp	3.4.1	5132
Filling	3.4.1	5132
Dimensional stability:		
Warp	3.5.1	5558
Filling	3.5.1	5558
Weave	3.4.2	Visual 2/
Creping after shrinkage	3.10	Visual 3/

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Table III. Test Methods (cont'd)

Characteristic	Requirement paragraph	Test method
Wrinkle recovery	3.10	4/
Pilling	3.10.1	5320 5/
pH	3.5.2	2811

1/ As specified in MIL-C-43655

2/ One determination per sample unit and the results reported as "pass" or "fail".

3/ Each shrinkage test specimen shall be visual examined for creping and the results reported as "pass" or "fail". Failure of any one of the test specimens shall be cause for rejection.

4/ The specimens shall be dry cleaned in accordance with Method 5580 of FED-STD-191, except that the specimens shall not be marked and evaluated for shrinkage and shall not be pressed. Wrinkle recovery shall be evaluated in accordance with AATCC Method 128-1969.

5/ The pilling test shall be conducted in accordance with Method 5320, except that five specimens of the test cloth and one specimen of the standard sample shall be used concurrently.

4.7.1 Polyester identification. The polyester fiber shall be dyed at the boil with Basic Red #14 using Orthophenylphenol as a carrier at a pH of 5 ±0.25 for 1 hour. Known samples of both the homopolymer type polyester and modified polyester shall be dyed simultaneously as a guide. The modified polymer will dye to a red shade, the homopolymer will show not more than a slight stain. This test is applied to the undyed fiber.

4.7.1.1 Fiber. The polyester filament shall be insoluble in each of the following solvents: acetone (boiling), 70 percent sulfuric acid, 5.0 percent sodium hydroxide (boiling), and methylene chloride (boiling). The methylene chloride shall be prepared by mixing nine parts reagent methylene chloride with one part U. S. P. (95 percent) ethyl alcohol. The methylene chloride solution is extremely toxic and adequate ventilation or use of fume hood is recommended. Unless otherwise specified, the above solvents shall be used at room temperature, using 100 times as much reagent as fiber. The fiber shall remain in the solvent for 10 minutes with intermittent stirring. This is a visual qualitative test.

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5. PACKAGING

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put up and packaged in accordance with the applicable requirements of PPP-P-1132 except that the moth repellent requirements shall not apply.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for Marine Corps women's maternity uniforms.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) When first article is required (see 3.2).
(c) Length required if other than specified (see 3.6).
(d) Width of cloth required when other than specified (see 3.7).
(e) Color of cloth required (see 3.8).
(f) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903, and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps, the cognizant procuring activity will provide comparable Data Item Descriptions. When so required, such data will be specified for delivery on a Form DD-1423 included in the contract.

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

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6.4 Dyestuff formulations. Suggested, but not mandatory, dyestuff formulations are as follows:

Marine Corps Green Shade 2243

Disperse Dyes (Polyester)	Acid Dyes (Wool)
Resolin Blue GFL	Acid Black 58
Foron Red 2G	Acid Red 174
Disperse Orange 41	Acid Yellow 114

6.5 Notes of caution to manufacturers.

6.5.1 Polyester fiber type. The choice of the polyester fiber type should be made by the contractor for a minimum pilling tendency and the fiber which will meet the colorfastness level of the specification (see 3.8.2).

6.5.2 Fiber length. The longest staple length commensurate with the staple of the wool, with proper twist multipliers to prevent pilling, should be used in the manufacture of this cloth (see 3.3.1.2).

6.5.3 Twist in the yarns. The twist in the yarns for the ply shall be such that the polyester fibers shall be so locked within the yarn structure that migration in subsequent processing and in the weaving of the cloth will not form pills (see 3.3.2).

6.5.4 Singeing. The cloth should be closely singed in order to minimize the amount of pilling (see 3.10.1).

6.5.5 Wrinkling. The requirements for fulling of the cloth and the balanced construction have been found to be essential to provide a cloth that will perform without substantial wrinkling and mussing in wear. Strict conformance to the fulling requirement is considered to be essential to adequate performance of the cloth (see 4.7).

6.6 Sears Fabric Defect Scales are available from:

Sears, Roebuck and Company
Department 733Q
"Sears Defect Replica Kit"
Sears Tower
Chicago, IL 60684

6.7 First article. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

Preparing activity:
Navy - MC
Project No. 8305-0705

Custodian:
Navy - MC

Review Activities:
Army - GL

User Activities
Navy - SA, CG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-29374 Cloth, Serge: Polyester/Wool

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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